

1 Exercise Sheet

1. uhhh... I'll pass on this one, chief. :/
2. A child has 12 blocks, of which 6 are black, 4 are red, 1 is white, and 1 is blue. If the child puts the blocks in a line, how many arrangements are possible?

$$\frac{12!}{6!4!1!1!}$$

3. How many different letter arrangements can be made from the letters MISSISSIPPI?

$$\frac{11!}{4!4!2!}$$

4. 20 workers are to be assigned to 20 different jobs, one to each job. How many different assignments are possible?

$$20!$$

5. In how many ways can 8 people be seated in a row if

(a) there are no restrictions on the seating arrangement?

$$8!$$

(b) persons A and B must sit next to each other?

$$2 \cdot 7!$$

(c) there are 4 men and 4 women and no 2 men or 2 women can sit next to each other?

$$2 \cdot 4! \cdot 4!$$

(d) there are 5 men and they must sit next to one another?

$$5! \cdot 4!$$

(e) there are 4 married couples and each couple must sit together?

$$4! \cdot (2!)^4$$

6. Letters in the Morse code are formed by a succession of dashes and dots with repetitions permitted. How many letters is it possible to form with ten symbols or less?

$$\sum_{k=1}^{10} 2^k = 2^{11} - 2$$

7. Consider a grid with points $(m, n) \in \mathbb{Z}^2$. Suppose you start at $(0, 0)$ and at each step you can go either one step up or one step to the right. How many paths are there from $(0, 0)$ to $(100, 100)$? How many paths from $(0, 0)$ to $(100, 100)$ pass through $(60, 40)$?

So we have to go right 100 times and up 100 times. So the number of paths is some combination of

r r r ... r u u u ... u

This we obviously know is just $\binom{200}{100}$.

In second one, we first go to $(60, 40)$ and then go from $(60, 40)$ to $(100, 100)$. So the number of paths is $\binom{100}{60}^2$.

8. If 8 identical blackboards are to be divided among 4 schools, how many divisions are possible?

$$\boxed{\binom{11}{3}}$$

How many if each school must receive at least 1 blackboard?

$$\boxed{\binom{7}{3}}$$

9. Suppose that 10 fish are caught at a lake that contains 5 distinct types of fish.

- (a) How many different outcomes are possible, where an outcome specifies the numbers of caught fish of each of the 5 types?

$$\boxed{\binom{14}{4}}$$

- (b) How many outcomes are possible when 3 of the 10 fish caught are trout?

$$\boxed{\binom{10}{3}}$$

- (c) How many when at least 2 of the 10 are trout?

$$\boxed{\binom{12}{4}}$$

10. If r_1 indistinguishable things of one kind and r_2 indistinguishable things of a second kind are placed into n cells, find the number of distinguishable arrangements.

$$\boxed{\binom{n+r_1-1}{r_1} \cdot \binom{n+r_2-1}{r_2}}$$

2 Exercise Sheet

1. $> \omega <$ About to cook mf